REVIEW · REVIEW

Week 1 recall day

Practice questions · Day 09 · Sun 1 Nov · pack R1

Part A — Machines 2024 mid-sem test (timed, 60 min)

Q1 Fill-in-the-Blank Questions [4]

2024 test Q1 (4 marks). *1) Fill-in-the-Blank Questions [4 marks]*

a) In the United States, the mains voltage is nominally 120 V, 60 Hz. This results in a peak AC voltage of _____ V and a rms AC voltage of _____V. The period of the mains frequency is equal to _____ ms. Thus, a phase shift of 90° corresponds to a phase shift of _____ ms.

b) A three-phase resistive load is connected in star and supplied by a 3ph 400V power supply and draws a line current of 15A. This means the line voltage is _____ V, the phase voltage is _____ V, the line current is _____ A and the phase current is _____ A.

Q2 Real and Reactive Power [4]

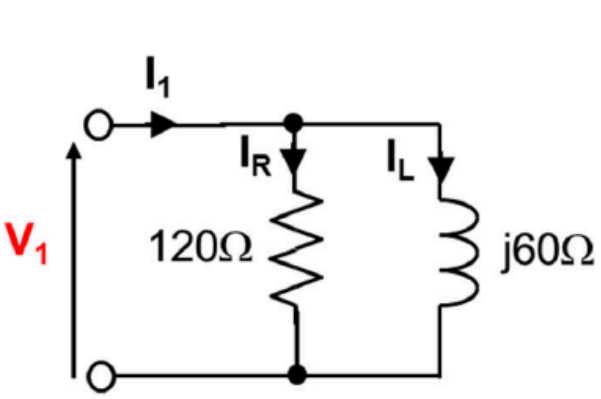
2024 test Q2 (4 marks). *2) Real and Reactive Power [4 marks]* A load is absorbing a real power of 1,200W and a reactive power of 400VAr.

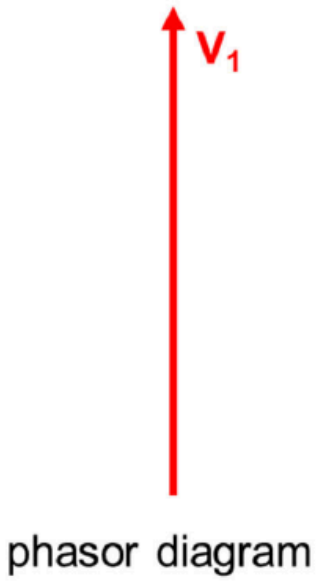
a) Find the complex power S in both cartesian and polar form. Find also the apparent power. [2 marks]

b) Sketch the power triangle and find the power-factor. [2 marks]

Q3 AC Circuit Analysis, parallel RL [8]

2024 test Q3 (8 marks). *3) AC Circuit Analysis [8 marks]* Consider the parallel resistor-inductor (RL) circuit shown below. It is connected to a single-phase 240V, 50Hz AC supply.





2024 test Q3 figure: 120 Ω in parallel with j60 Ω across V1, with I1, IR, IL marked, and the phasor-diagram template with V1 drawn vertically (from the test paper)

a) Find the following phasor quantities: supply voltage $\mathbf{V}_1$, resistor current $\mathbf{I}_R$, inductor current $\mathbf{I}_L$ and supply current $\mathbf{I}_1$ in polar form. [4 marks]

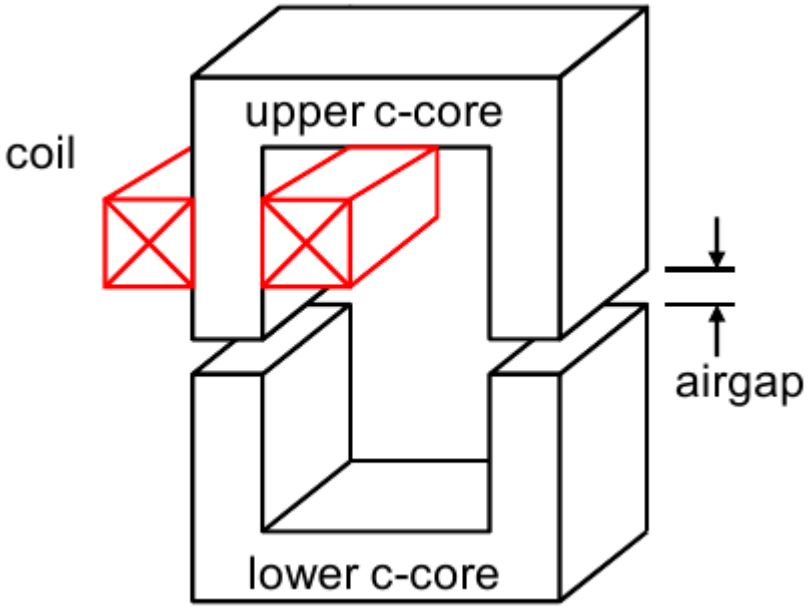
b) Using the template above showing $\mathbf{V}_1$, neatly sketch the phasors $\mathbf{I}_R$, $\mathbf{I}_L$ and $\mathbf{I}_1$, clearly labelling each one. [2 marks]

c) Calculate the real and reactive power absorbed by the circuit. [2 marks]

The figure gives $R = 120\ \Omega$ and $jX_L = j60\ \Omega$ (the same values the official answers give).

Q4 Electromagnetics [7]

2024 test Q4 (7 marks). *4) Electromagnetics [7 marks]* An electromagnet is constructed using two identical magnetic cores as shown on the right. The upper c-core is held stationary and magnetic attraction is used to hold the lower c-core. Each core has a mass of 0.3 kg, a magnetic cross-section area A of 5 cm² and a relative permeability of 4,000. The mean magnetic iron path in the cores for the flux is 25 cm (this is the total for the two cores). A coil of five turns carrying 3.5 A is placed on the upper core.



2024 test Q4 figure: upper c-core with the coil, airgap, and lower c-core (from the test paper)

a) Assuming initially that the airgap between the cores is zero, what is the flux density in the magnetic circuit? [2 marks]

b) A piece of paper of 0.07mm thickness is placed between the cores. The relative permeability of paper can assumed to be equal to 1. Calculate the new value of flux density in the magnetic circuit. [3 marks]

c) What is the minimum flux density in the magnetic circuit that would be required to support the weight of the lower core? Assume a value of acceleration due to gravity of 9.81m/s². [2 marks]

Q5 Magnetic Circuits [5]

2024 test Q5 (5 marks). *5) Magnetic Circuits [5 marks]* A single-phase 240V/48V, 50 Hz transformer uses a laminated rectangular core with a cross-sectional area of 40 cm², magnetic path length of 0.8 m and relative permeability of 2,000. The peak operating flux density of the core should not exceed 1.6 T.

a) Using the back-emf equation, calculate the required number of primary turns for the transformer. [2 marks]

b) The primary winding consists of copper wire with a length of 40 m and a circular wire of diameter 0.5 mm. Find its resistance using a conductivity of $59 \times 10^6\ (\Omega\mathrm{m})^{-1}$. [2 marks]

c) Choose an appropriate number of turns for the secondary winding. [1 mark]

Q6 Transformer Parameters [4]

2024 test Q6 (4 marks). *6) Transformer Parameters [4 marks]* Consider a star-connected, three-phase transformer. The following test results were obtained:

- open-circuit test: 5.2 kV, 45 A, 49 kW
- short-circuit test: 710 V, 390 A, 140 kW

Find the winding resistance and the leakage reactance for the simplified transformer equivalent circuit. It is not necessary to find the other equivalent circuit parameters.

Q7 Transformer Analysis [4]

2024 test Q7 (4 marks). *7) Transformer Analysis [4 marks]* A three-phase, 14kV:135kV 250MVA transformer is supplying a load. The load has an apparent power of 150 MVA and is generating 35 MVAr of reactive power. The output voltage is equal to its rated value. Find the secondary phase voltage V_2' and phase current I_2' phasors referred to the primary. Assume the phase angle of V_2' is zero. Hint: remember the power triangle!

Q8 Short-Answer Question [4]

2024 test Q8 (4 marks). *8) Short-Answer Question [4 marks]* Answer the following question using words only (that is, without using sketches or equations). Consider an ideal transformer driving a resistive load of resistance R. Carefully explain what would be the effect on the output voltage and current if the load resistance was doubled. Carefully explain why and how this would be different for a real transformer.

Part A scoring

Add up your marks out of 40. Note which questions lost marks and which pack each maps to (see the table at the top of Part A). Bring that list to Part C.

Part B — Mixed recall (45 min)

B1. [D · D1] $x(t) = \cos(2\pi \cdot 700t)$ is sampled at $F_s = 1$ kHz. Give ω_0 reduced into $(-\pi, \pi]$, and the frequency an ideal reconstructor outputs.

B2. [M · M1] A single-phase load on 230 V draws 5 A at 0.8 pf lagging. Find P and Q, with the sign of Q.

B3. [E · E1] Si at 300 K is doped p-type with $N_A = 10^{16}\ \mathrm{cm}^{-3}$. What are p and n?

B4. [M · M4] **Method.** You are given open-circuit and short-circuit test data for a transformer. Which test gives which parameters, and which circuit model (series or parallel) do you use for each?

B5. [E · E3] A half-wave rectifier is fed from a 12 V rms transformer secondary. With the constant-voltage-drop model (Si, 0.7 V), what is the peak output voltage? What PIV must the diode withstand with no smoothing capacitor?

B6. [D · D1] Is $y(n) = n\,x(n)$ linear, time-invariant, causal and BIBO stable?

B7. [M · M2] A 2 kW load at 0.7 pf lagging is on 240 V, 50 Hz. What parallel capacitor corrects it to unity pf?

B8. [E · E2] A diode with $n = 1$ carries a current that rises by a factor of 4. By how much does V_D rise?

B9. [D · D2] Convolve $x(n) = \{\underline{1}, 2\}$ with $h(n) = \{\underline{1}, 1, 1\}$ (the underline marks $n = 0$).

B10. [M · M2] A balanced star-connected load on a 415 V supply draws 10 A line current at 0.85 pf lagging. Find the total P and Q.

B11. [E · E4] Fixed bias: $V_{CC} = 12$ V, $R_B = 470$ kΩ from V_{CC} to the base, $R_C = 2.2$ kΩ, $\beta = 100$, emitter grounded. Find I_B, I_C and V_{CE}, and confirm the operating mode.

B12. [D · D3] Give the DTFT of $x(n) = a^n u(n)$ and the condition for it to exist. For $a = 0.5$, what is $|X(\omega)|$ at $\omega = 0$ and at $\omega = \pi$?

B13. [M · M3] A coil of 200 turns is on a core with $l = 0.3$ m, $A = 4\ \text{cm}^2$ and $\mu_R = 2000$ (no gap). Find the reluctance and the inductance.

B14. [E · E3] **Method.** A Zener regulator has $V_S = 15$ V, a series resistor $R = 200\ \Omega$ and $V_Z = 6.2$ V (ideal Zener), with a 1 kΩ load. First, how do you check that the Zener is actually regulating? Then find I_Z.

B15. [D · D2] **Method.** $x(n) = (\frac{1}{2})^n u(n)$ is applied to $h(n) = (\frac{1}{3})^n u(n)$. What is the fastest route to $y(n)$? Give the closed form.

B16. [M · M4] A 2400 V/240 V single-phase transformer has a 2 Ω load on its secondary. What impedance does the supply see, referred to the primary (ideal transformer)?

B17. [E · E4] $V_{BB} = 5$ V drives the base through $R_B = 20$ kΩ. $V_{CC} = 10$ V, $R_C = 1$ kΩ, $\beta = 100$, emitter grounded, $V_{CE,sat} = 0.2$ V. Is the BJT active or saturated? Find I_C.

B18. [D · D4] $h(n) = 0.5\delta(n) + 0.5\delta(n-1)$. Find $H(\omega)$. What is the output for $x(n) = \cos(\pi n/2)$?

B19. [M · M5] A transformer has 20 W iron loss and 45 W copper loss at full load. At what fraction of full load is its efficiency highest?

B20. [E · E2] **Method.** A circuit has two diodes and you don't know which conduct. Describe the procedure.

B21. [D · D4] $y(n) = 0.9\,y(n-1) + x(n)$. Find $H(\omega)$ and its gain at $\omega = 0$ and $\omega = \pi$. What kind of filter is it?

B22. [M · M4] **Method.** Why does the OC test measure (nearly) only iron loss, and the SC test (nearly) only copper loss?

B23. [D · D3] $X(\omega)$ is the DTFT of $x(n)$. Write the DTFT of $x(n-3)$ and of $(-1)^n x(n)$. What does the second one do to a lowpass spectrum?

B24. [E · E3] A clipper: the input $v_i = 5 \sin \omega t$ V is fed through a series resistor to the output. Across the output (output to ground) is a Si diode (0.7 V model) in series with a 2 V battery. The diode's anode is at the output, its cathode goes to the battery's + terminal, and the battery's − terminal goes to ground. What does the output look like?

Part B scoring

Count your ✓, ~ and × for each subject:

Subject	Questions	✓	~	×
Machines	B2, B4, B7, B10, B13, B16, B19, B22			
DSP	B1, B6, B9, B12, B15, B18, B21, B23			
EDC	B3, B5, B8, B11, B14, B17, B20, B24			